

ICT702 Data Visualisation

Exploring Data Visually

Module 6



I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



Part 3

Types of Missing Data

Missing data are deemed **legitimate** when they naturally occur.

- Example: a survey question following a N/A answer.

When the missing data do not occur naturally, they are deemed **illegitimate**.

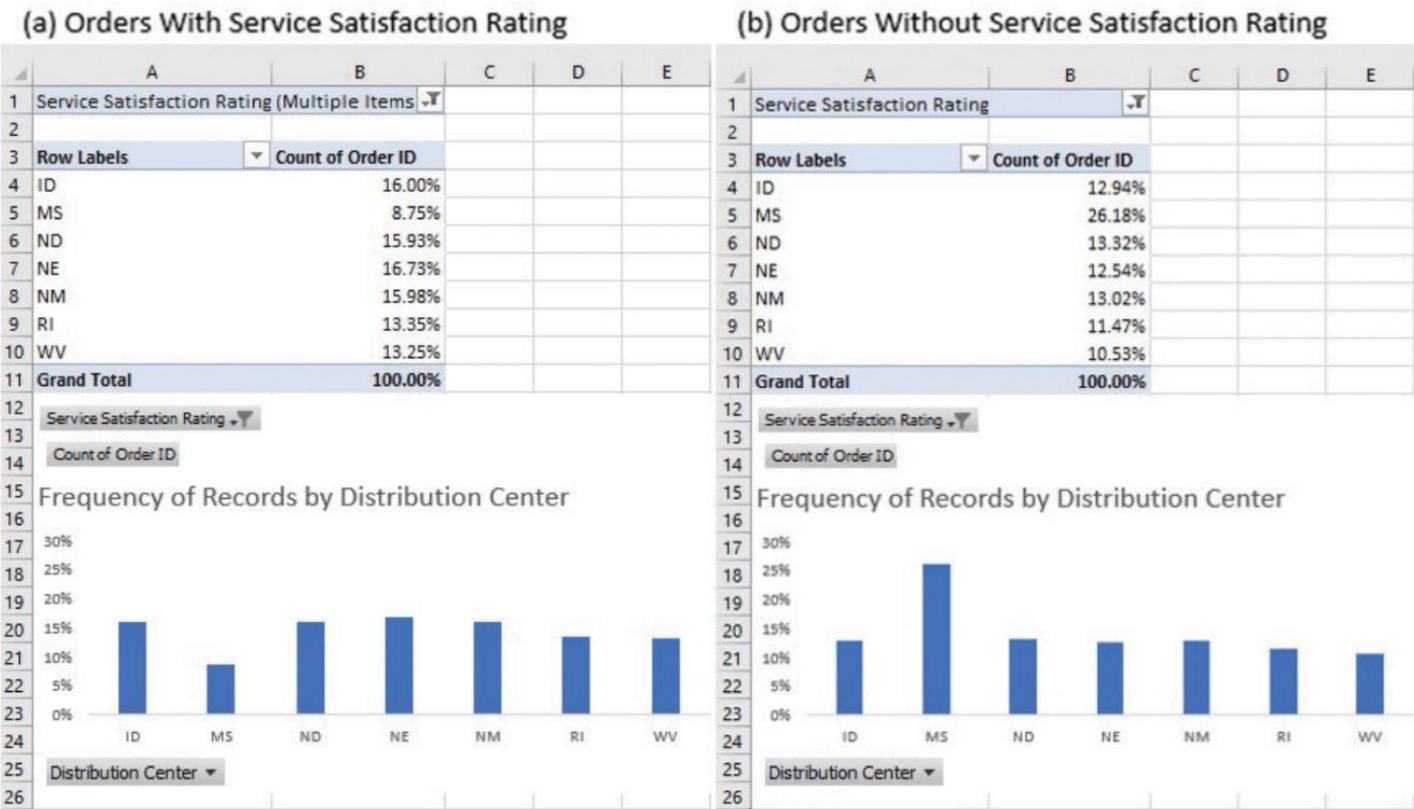
Options available to address illegitimately missing data:

- Discard observations (rows) with missing values.
- Fill in missing entries with estimated values.
- Treat missing data as a separate category.

Types of missing values:

- **MCAR** – Missing Completely at Random (not related to any variable or variable values)
- **MAR** – Missing at Random (missing values for a variable related to another variable)
- **MNAR** – Missing Not at Random (missing specific values for a variable)

Exploring Patterns Associated with Missing Data



Comparing Frequency for Records by Distribution Center (a) with vs. (b) without a Service Satisfaction Rating in the *EJB* data



Visualizing Time Series Data

Time series data consists of observations collected at different points in time.

Cross-sectional data consists of observations collected at a single point in time.

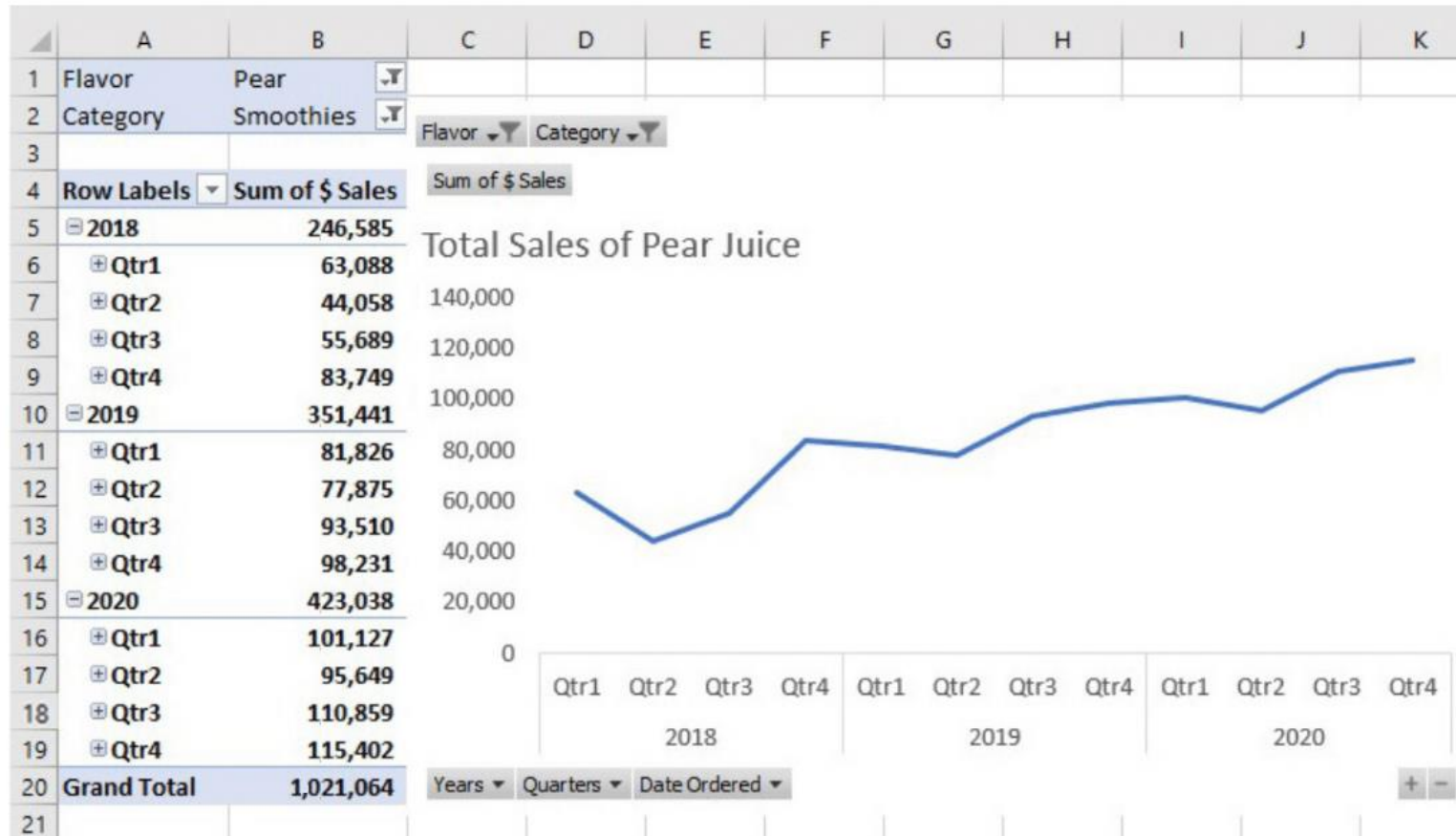
Time series data are often displayed using **line charts**.

Line charts allow to distinguish the following features of time series data:

- **Trend:** the long-run pattern in values observable over several time periods.
- **Variability:** the difference in values from period to period.
- **Seasonality:** a pattern in time series data that recurs periodically over a regular time interval.

Viewing Data at Different Temporal Frequencies

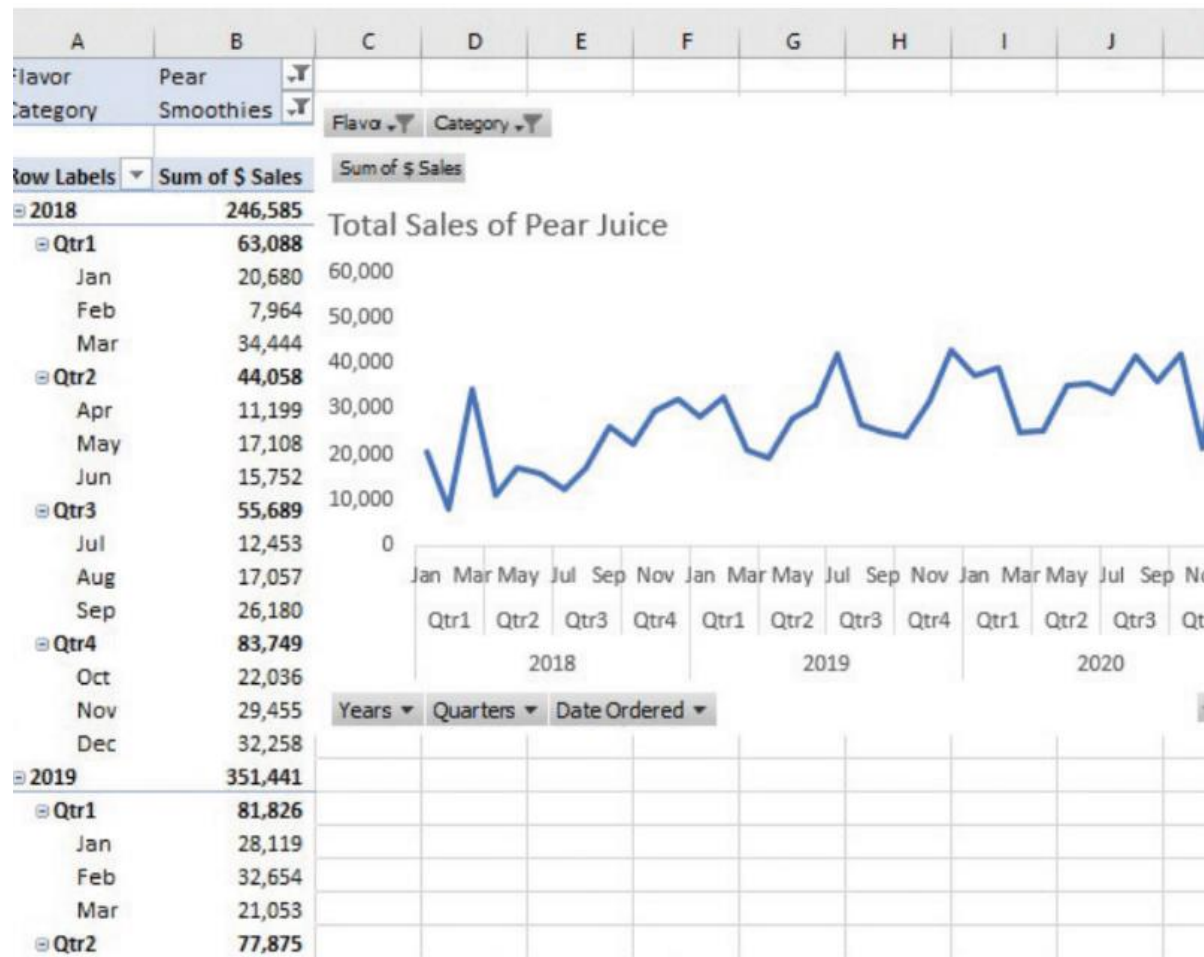
PivotTable and PivotChart of quarterly pear juice sales in the *EJB* data



The **temporal frequency** is the rate at which the data is displayed on the horizontal axis of a time series chart.

Effect of Temporal Frequencies on Visualisation

PivotTable and PivotChart of monthly pear juice sales in the *EJB* data



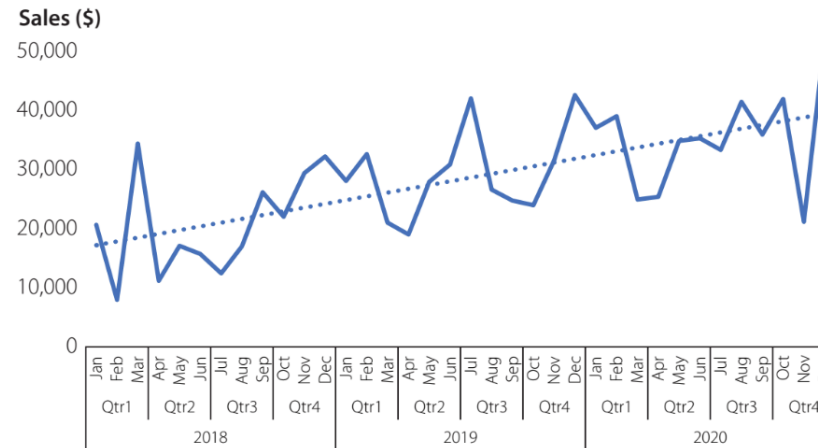
The pear juice monthly sales data to the left suggest a less clear seasonality pattern across the twelve months.



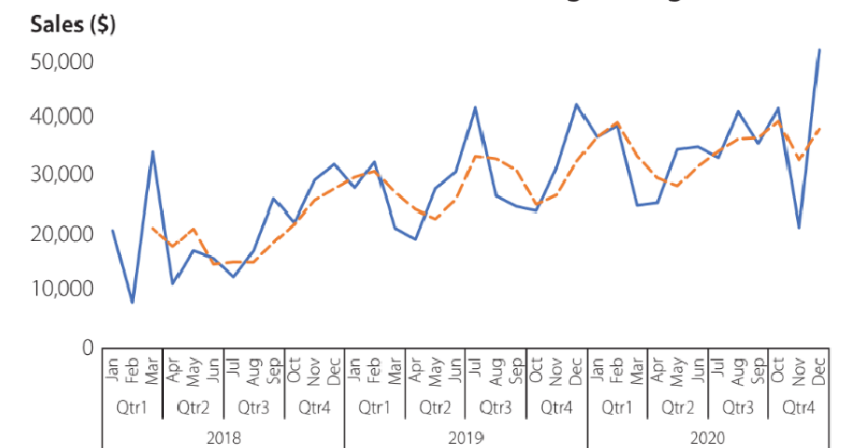
Patterns in Time Series Data: Trellis Display

Trellis display of pear juice sales showing linear trendline, 3-month, 6-month, and 12-month moving averages

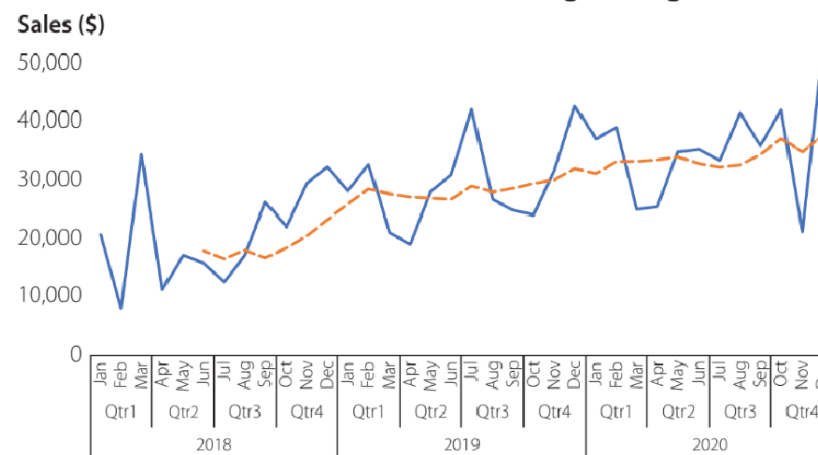
Total Sales of Pear Juice



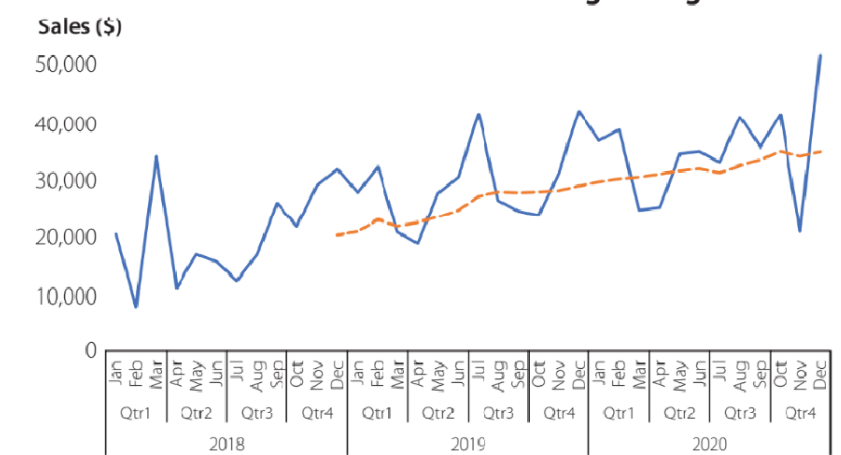
Total Sales of Pear Juice: 3-Month Moving Average



Total Sales of Pear Juice: 6-Month Moving Average



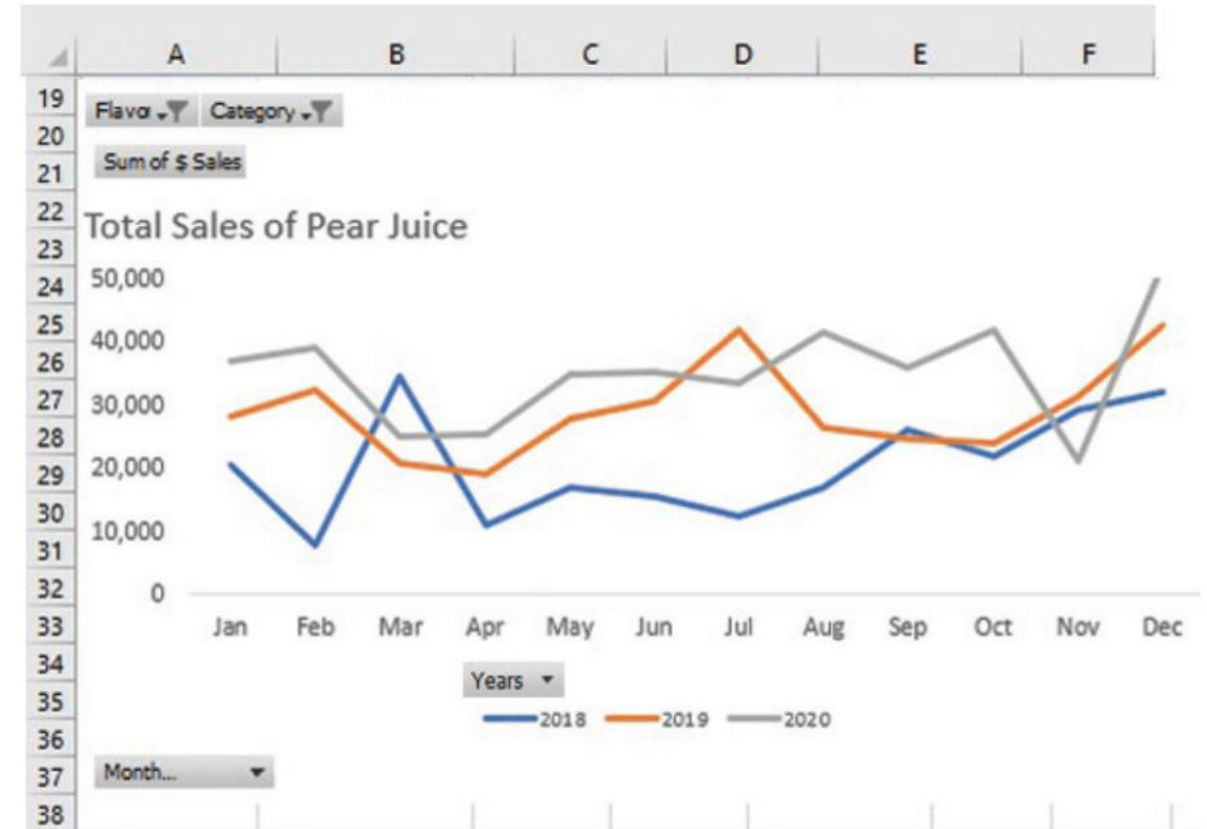
Total Sales of Pear Juice: 12-Month Moving Average



Exploring Seasonality in Time Series Data

Exploring pear juice sales in the *EJB* data for monthly seasonality

	A	B	C	D	E	F
1	Flavor	Pear				
2	Category	Smoothies				
3						
4	Sum of \$ Sales	Column Labels				
5	Row Labels	2018	2019	2020	Grand Total	
6	Jan	20,680	28,119	37,097	85,896	
7	Feb	7,964	32,654	39,079	79,697	
8	Mar	34,444	21,053	24,951	80,448	
9	Apr	11,199	19,030	25,416	55,645	
10	May	17,108	27,948	34,899	79,955	
11	Jun	15,752	30,897	35,334	81,983	
12	Jul	12,453	42,095	33,384	87,932	
13	Aug	17,057	26,646	41,515	85,218	
14	Sep	26,180	24,769	35,960	86,908	
15	Oct	22,036	24,002	41,977	88,015	
16	Nov	29,455	31,587	21,180	82,222	
17	Dec	32,258	42,641	52,245	1,27,144	
18	Grand Total	2,46,585	3,51,441	4,23,038	10,21,064	



Rearranging Data for Visualisation

NBA3PA data in (a) stacked and (b) unstacked form

(a) Stacked Data

	A	B	C
1	Team	Year	Three Point Attempts
2	Atlanta Hawks	2015	2152
3	Boston Celtics	2015	2021
4	Brooklyn Nets	2015	1633
5	Charlotte Hornets	2015	1566
6	Chicago Bulls	2015	1825

(b) Unstacked Data

	A	B	C	D	E	F
1		Three Point Attempts				
2	Team	2015	2016	2017	2018	2019
3	Atlanta Hawks	2152	2326	2137	2544	3034
4	Boston Celtics	2021	2142	2742	2492	2829
5	Brooklyn Nets	1633	1508	2591	2924	2965
6	Charlotte Hornets	1566	2410	2347	2233	2783
7	Chicago Bulls	1825	1753	1831	2549	2123

Two major data arrangements:
Stacked (or single column) data
Unstacked (or split) data

Rearranging Data for Visualisation: Sparklines

Add sparklines to the unstacked three-point shot attempt data

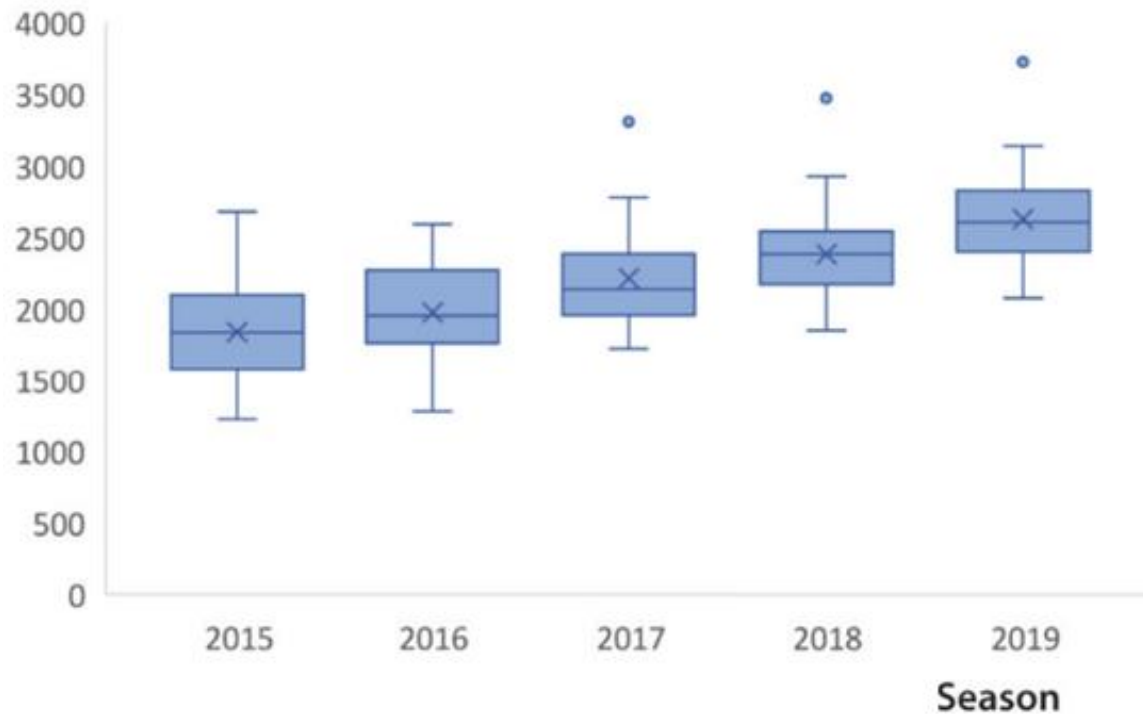
	A	B	C	D	E	F	G
1		Three Point Attempts					
2	Team	2015	2016	2017	2018	2019	
3	Milwaukee Bucks	1500	1277	1946	2024	3134	
4	Golden State Warriors	2217	2592	2562	2370	2824	
5	New Orleans Pelicans	1583	1951	2196	2312	2449	
6	Philadelphia 76ers	2160	2255	2443	2445	2474	
7	Los Angeles Clippers	2202	2190	2245	2196	2118	
8	Portland Trail Blazers	2231	2336	2272	2308	2520	
9	Oklahoma City Thunder	1864	1945	2116	2491	2677	
10	Toronto Raptors	2060	1915	1996	2705	2771	
11	Sacramento Kings	1350	1839	1960	1967	2455	
12	Washington Wizards	1381	1983	2030	2173	2731	
13	Houston Rockets	2680	2533	3306	3470	3721	
14	Atlanta Hawks	2152	2326	2137	2544	3034	
15	Minnesota Timberwolves	1223	1347	1723	1845	2357	
16	Boston Celtics	2021	2142	2742	2492	2829	
17	Brooklyn Nets	1633	1508	2591	2924	2965	
18	Los Angeles Lakers	1546	2016	2110	2384	2541	
19	Utah Jazz	1781	1956	2128	2425	2789	
20	San Antonio Spurs	1847	1518	1927	1977	2071	
21	Charlotte Hornets	1566	2410	2347	2233	2783	
22	Denver Nuggets	2032	1943	2365	2536	2571	
23	Dallas Mavericks	2082	2342	2473	2688	3002	
24	Indiana Pacers	1740	1889	1885	2010	2081	
25	Phoenix Suns	2048	2118	1854	2286	2400	
26	Orlando Magic	1598	1818	2139	2405	2633	
27	Detroit Pistons	2043	2148	1915	2373	2854	
28	Miami Heat	1659	1480	2213	2506	2658	
29	Chicago Bulls	1825	1753	1831	2549	2123	
30	New York Knicks	1614	1762	2022	1914	2421	
31	Cleveland Cavaliers	2253	2428	2779	2636	2388	
32	Memphis Grizzlies	1246	1521	2169	2152	2368	

A **sparkline** is a minimalist type of line chart directly placed into a spreadsheet cell.

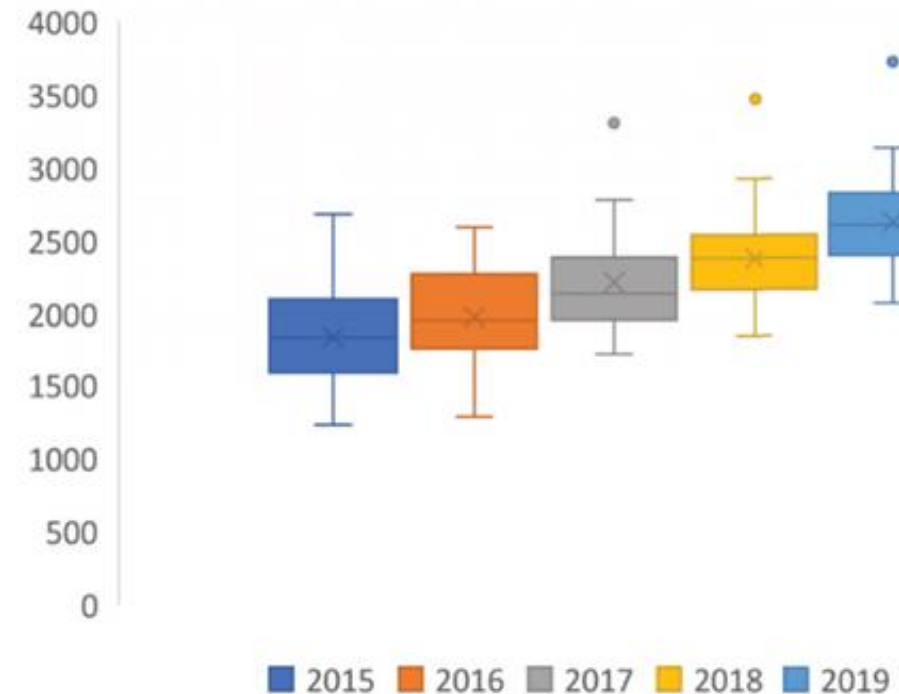
Rearranging Data in Box and Whisker Plots

Box and whisker plots built from stacked and unstacked three-point shot attempt data

NBA Teams Shooting More Three Pointers
Three-Point Attempts



NBA Teams Shooting More Three Pointers
Three-Point Attempts



End of Part 3

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